

Question 1

15 marks

Start a new page

MARKS

(a) If $f(x) = (x-1)(x-3)$ then sketch

(i) $y = \frac{1}{f(x)}$

2

(ii) $y = f(|x|)$

2

(iii) $|y| = f(x)$

2

(b) (i) Find the stationary points and the asymptotes of the function

2

$$y = \frac{(x+1)^4}{x^4+1}$$

(ii) Sketch this function labelling all essential features.

1

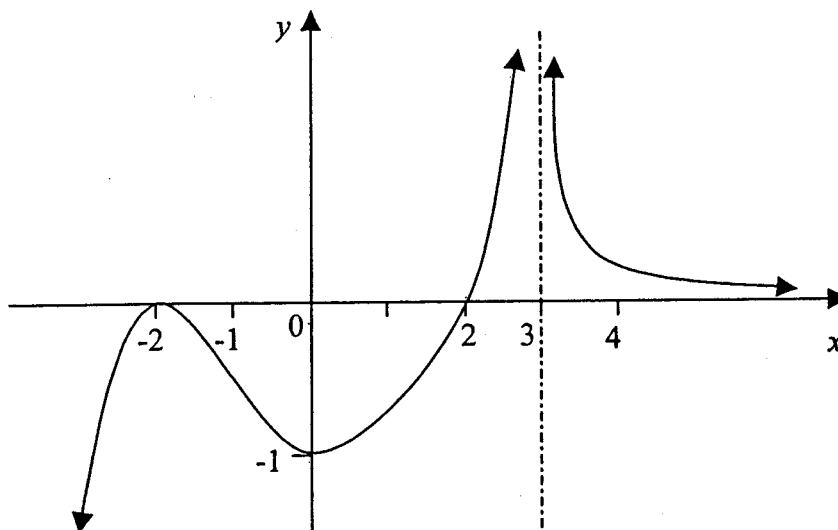
(iii) Use the graph to find the set of values of k for which

2

$$(x+1)^4 = k(x^4+1)$$
 has two distinct real roots.

(c) Given the graph of $y = f'(x)$ below, sketch the graph of $y = f(x)$.

4

 $y = f'(x)$ is the derivative of $y = f(x)$.

Question 2

15 marks

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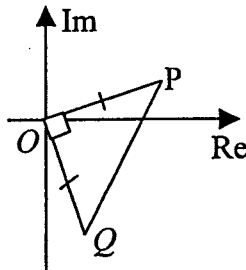
MARKS

- (a) (i) Find $\int \frac{x}{\sqrt{9-16x^2}} dx$ 2
- (ii) Find $\int \frac{x^2}{x+1} dx$ 2
- (iii) Evaluate $\int_0^{\ln 3} xe^x dx$ 3
- (b) (i) Find real numbers A, B and C such that $\frac{2}{(t+1)(t^2+1)} = \frac{A}{t+1} + \frac{Bt+C}{t^2+1}$ 3
- (ii) Hence, find $\int_0^1 \frac{2}{(t+1)(t^2+1)} dt$. 3
- (iii) By using the substitution $t = \tan\left(\frac{x}{2}\right)$ evaluate $\int_0^{\frac{\pi}{2}} \frac{\sin x}{1 + \sin x - \cos x} dx$. 2

Question 3

15 marks

Start a new page

- (a) Evaluate $\arg((2+i)\bar{w})$ given $w = -1-3i$. 2
- (b) Write $x^2 - 12x + 48$ as the product of two linear factors. 2
- (c) In the diagram on the right, triangle POQ is right-angled and isosceles. If P represents the complex number $a+bi$, where a and b are real, find the complex number represented by Q . 1
- 
- (d) Sketch in the Argand diagram the locus of the complex number z given:
- (i) $\arg(z-2) = \arg z + \frac{\pi}{2}$ 2
- (ii) $|z+3i| < 2|z|$ 3
- (e) (i) Find, in modulus-argument form, the three cube roots of -8 . 2
- (ii) Write the two unreal cube roots of -8 in the form $a+bi$, where a and b are real. 1
- (iii) If w_1 and w_2 are the the unreal cube roots of -8 , show that 2
- $$w_1^{6n} + w_2^{6n} = 2^{6n+1} \text{ for all integers } n.$$

Question 4

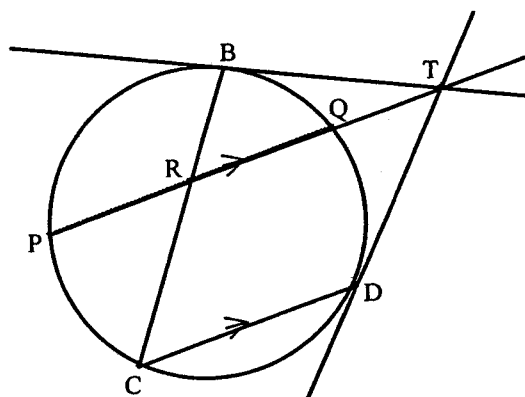
15 Marks

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MARKS

- (a) Factorise $P(x) = x^4 - 5x^3 + 4x^2 + 2x - 8$ over
- (i) Real numbers; 2
- (ii) Complex numbers. 1
- (b) Write down all polynomials that have degree 4 with 3 as a single zero and -1 as a zero of multiplicity 3. 1
- (c) If α, β, γ are the roots of $x^3 - 2x^2 + x + 3 = 0$, evaluate:
- (i) $\alpha^2 + \beta^2 + \gamma^2$ 2
- (ii) $\alpha^3 + \beta^3 + \gamma^3$ 1
- (d) If α, β, γ are the roots of $x^3 + 2x^2 - 2x + 3 = 0$, form the equation whose roots are:
- (i) $2\alpha, 2\beta, 2\gamma$ 1
- (ii) $\alpha^2, \beta^2, \gamma^2$. 3

(e)



In the diagram, PQ and CD are parallel chords of a circle. The tangent at D meets PQ produced externally at T . B is the point of contact of the other tangent from T to the circle. BC meets PQ internally at R .

Copy or trace this diagram onto your answer page.

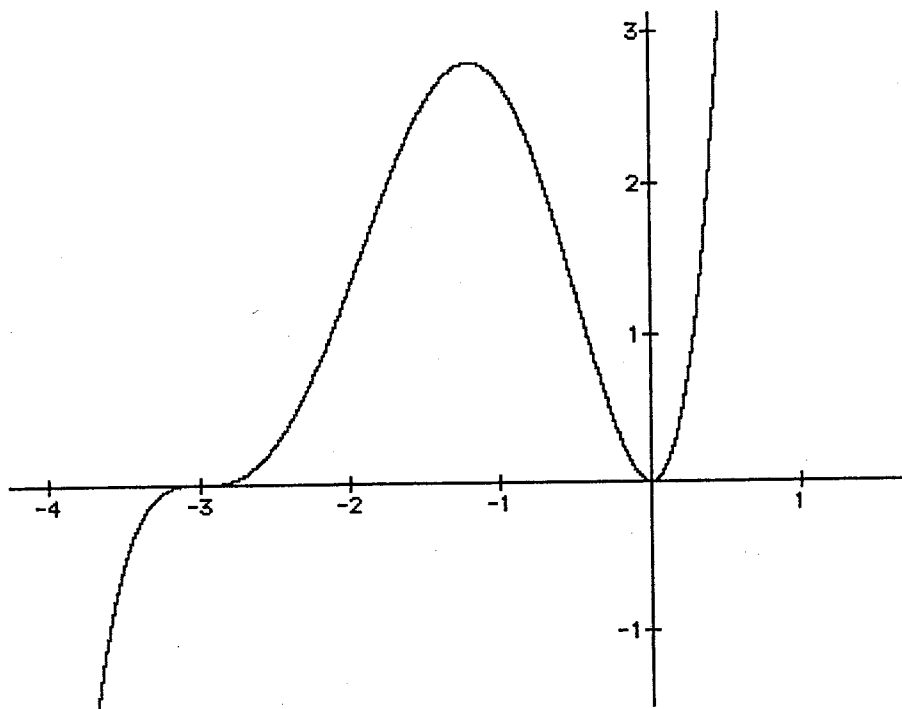
- (i) Explain why $\angle BDT = \angle BRT$. 2
- (ii) Show that B, T, D and R are concyclic points. 2

Question 5

15 Marks

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MARKS



- (a) Consider the graph of $y = f(x)$ as shown above.
On the answer sheet provided, use the graphs of $y = f(x)$ to clearly sketch separately the graphs of:
- (i) $y = \frac{1}{f(x)}$ 2
- (ii) $y^2 = f(x)$ 2
- (iii) $y = f'(x)$ 1
- (b) Suggest a possible polynomial equation for the graph of $y = f(x)$ shown in part (a) of Q5. 1
- (c) A solid S is formed by rotating the region bounded by the parabola $y^2 = 16(1-x)$ and the y axis through 360° about the line $x = 2$.
- (i) By slicing perpendicular to the axis of rotation, find the exact volume of S . 4
- (ii) (α) Use the method of cylindrical shells to show that the volume of S is 2
also given by $\int_0^1 16\pi(2-x)\sqrt{1-x} \, dx$.
- (β) Confirm your answer to part (i) by calculating this definite integral using the substitution $u = 1-x$ 3

Question 6

15 Marks

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MARKS

- (a) A solid has as its base the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$. 4
 If each section perpendicular to the major axis is an equilateral triangle, show that the volume of the solid is $128\sqrt{3}$ units³.
- (b) The region $(x - 2R)^2 + y^2 \leq R^2$ is rotated about the y -axis forming a solid of revolution called a torus. 6
 By summing volumes of cylindrical shells, show that the volume of the torus is $4\pi^2 R^3$ units³.
- (c) The angles of elevation of the top of a tower P from three points A, B, C are α, β, γ respectively. 2
 A, B, C are in a straight line such that $AB = BC = a$, but the line AC does not pass through S , the base of the tower.
- (i) If $\angle ABS = \theta$ and h is the height of the tower, show that 2
 $CS^2 = a^2 + h^2 \cot^2 \beta + 2ah \cot \beta \cos \theta$.
- (ii) Prove that the height of the tower is $\frac{a\sqrt{2}}{\left\{ \cot^2 \alpha + \cot^2 \gamma - 2 \cot^2 \beta \right\}^{\frac{1}{2}}}$. 3

Question 7

15 Marks

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- (a) $P\left(3p, \frac{3}{p}\right)$ and $Q\left(3q, \frac{3}{q}\right)$ are points on the rectangular hyperbola $xy = 9$. 1
 The equation of chord PQ is $x + pqy = 3(p + q)$.
- (i) Find the co-ordinates of N , the midpoint of PQ . 1
- (ii) If the chord PQ is a tangent to the parabola $y^2 = 3x$, prove that the locus of N is $3x = -8y^2$. 3
- (b) The equation of a curve is $x^2 y^2 - x^2 + y^2 = 0$. 2
- (i) Show that the numerical value of y is always less than 1. 1
- (ii) Find the equations of the vertical asymptotes. 1
- (iii) Show that $\frac{dy}{dx} = \frac{y^3}{x^3}$. 3
- (iv) Sketch the curve. 1

Question 7 continues on next page.

Question 7 continued

MARKS

- (c) A ball thrown from a point P with velocity V , at an inclination α to the horizontal, reaches a point Q after t seconds.

Show that if PQ is inclined at θ to the horizontal, (where $\alpha > \theta$), then the direction of motion of the ball, when at Q , is inclined to the horizontal at an acute angle of $\tan^{-1}[2 \tan \theta - \tan \alpha]$.

You may use the result without proof

$$x = V \cos \alpha \times t$$

$$y = V \sin \alpha \times t - \frac{1}{2} g t^2$$

4

Question 8

15 Marks

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- (a) Let $I_n = \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \operatorname{cosec}^n x \, dx$ where n is a positive integer.

- (i) Using integration, show that $(n-1)I_n = 2^{n-2}\sqrt{3} + (n-2)I_{n-2}$.

4

- (ii) Evaluate $J = \int_0^{\frac{\pi}{3}} \sec^4 x \, dx$.

3

- (b) Consider the polynomial $x^5 - i = 0$.

- (i) Show that $1 - ix - x^2 + ix^3 + x^4 = 0$ for $x \neq i$.

2

- (ii) Show that $(x-i)\left(x^2 - 2i \sin \frac{\pi}{10} x - 1\right)\left(x^2 + 2i \sin \frac{3\pi}{10} x - 1\right) = 0$.

4

- (iii) Show that $\sin \frac{\pi}{10} \sin \frac{3\pi}{10} = \frac{1}{4}$.

2

Question 8 continued

(b) (i)

$$x^5 - i = (x - i)(x^4 + ix^3 + i^2x^2 + i^3x + i^4)$$

$$= (x - i)(x^4 + ix^3 - x^2 - ix + 1) = 0$$

$$x \neq i \quad \text{so} \quad 1 - ix - x^2 + ix^3 + x^4 = 0$$

(ii) $x^5 - i = 0 \Rightarrow x^5 = i \quad (rcis\theta)^5 = cis\frac{\pi}{2}$

$$r = 1, \quad cis5\theta = cis\frac{\pi}{2}$$

$$5\theta = \frac{\pi}{2} + 2k\pi$$

$$\theta = \frac{\pi}{10} + \frac{2k\pi}{5}, \quad k = 0, 1, 2, 3, 4$$

$$\theta = \frac{\pi}{10}, \frac{\pi}{2}, \frac{9\pi}{10}, \frac{13\pi}{10}, \frac{17\pi}{10} \quad \text{or} \quad \theta = \frac{\pi}{10}, \frac{\pi}{2}, \frac{9\pi}{10}, \frac{-3\pi}{10}, \frac{-7\pi}{10}$$

$$(x - i)\left(x - cis\frac{\pi}{10}\right)\left(x - cis\frac{9\pi}{10}\right)\left(x - cis\frac{-3\pi}{10}\right)\left(x - cis\frac{-7\pi}{10}\right) = 0$$

$$\text{Now} \quad cis\frac{\pi}{10} + cis\frac{9\pi}{10} = \cos\frac{\pi}{10} + i\sin\frac{\pi}{10} + \cos\frac{9\pi}{10} + i\sin\frac{9\pi}{10}$$

$$= \cos\frac{\pi}{10} + i\sin\frac{\pi}{10} - \cos\frac{\pi}{10} + i\sin\frac{\pi}{10} = 2i\sin\frac{\pi}{10}$$

$$\text{and} \quad cis\frac{\pi}{10} \times cis\frac{9\pi}{10} = \left(\cos\frac{\pi}{10} + i\sin\frac{\pi}{10}\right)\left(\cos\frac{9\pi}{10} + i\sin\frac{9\pi}{10}\right)$$

$$= \left(\cos\frac{\pi}{10} + i\sin\frac{\pi}{10}\right)\left(-\cos\frac{\pi}{10} + i\sin\frac{\pi}{10}\right)$$

$$= -\cos^2\frac{\pi}{10} - \sin^2\frac{\pi}{10} = -1$$

$$\text{Similarly} \quad cis\frac{-3\pi}{10} + cis\frac{-7\pi}{10} = 2i\sin\frac{-3\pi}{10} = -2i\sin\frac{3\pi}{10}$$

$$\text{and} \quad cis\frac{\pi}{10} \times cis\frac{9\pi}{10} = -1$$

$$\text{Hence} \quad (x - i)\left(x^2 - 2i\sin\frac{\pi}{10}x - 1\right)\left(x^2 + 2i\sin\frac{3\pi}{10}x - 1\right) = 0$$

(iii) From (i) and (ii) you can write:

$$\left(x^2 - 2i\sin\frac{\pi}{10}x - 1\right)\left(x^2 + 2i\sin\frac{3\pi}{10}x - 1\right) = x^4 + ix^3 - x^2 - ix + 1$$

The coefficient of x^2 will involve the product $\sin\frac{\pi}{10}\sin\frac{3\pi}{10}$.

$$\text{ie.} \quad -x^2 - 4i^2\sin\frac{\pi}{10}\sin\frac{3\pi}{10}x^2 - x^2 = -x^2$$

$$4\sin\frac{\pi}{10}\sin\frac{3\pi}{10} = 1$$

$$\sin\frac{\pi}{10}\sin\frac{3\pi}{10} = \frac{1}{4}$$

2

Evidence of the factorisation with powers of i needed for 2 marks

4

1

1

1

1 for these two lines from similarly

2

1 for this line with evidence of source

1 penultimate line

	$x = Vt \cos \alpha$ $y = Vt \sin \alpha - \frac{gt^2}{2}$ $P(0,0), \quad Q \left(Vt \cos \alpha, Vt \sin \alpha - \frac{gt^2}{2} \right)$ Gradient of PQ = $\frac{Vt \sin \alpha - \frac{gt^2}{2}}{Vt \cos \alpha} = \tan \alpha - \frac{gt}{2V \cos \alpha}$ Hence $\tan \theta = \tan \alpha - \frac{gt}{2V \cos \alpha}$ At Q, $\dot{x} = V \cos \alpha$ $\frac{dy}{dx} = \frac{V \sin \alpha - gt}{V \cos \alpha}$ $\tan \beta = \tan \alpha - \frac{gt}{V \cos \alpha}$ so $\frac{gt}{V \cos \alpha} = \tan \alpha - \tan \beta$ Hence $\tan \theta = \tan \alpha - \frac{\tan \alpha - \tan \beta}{2}$ $2 \tan \theta = 2 \tan \alpha - \tan \alpha - \tan \beta$ $\tan \beta = 2 \tan \theta - \tan \alpha$ $\beta = \tan^{-1}(2 \tan \theta - \tan \alpha)$	4 1 for correct diagram if nothing else with no multiple use of θ
Gradient of PQ = $\frac{Vt \sin \alpha - \frac{gt^2}{2}}{Vt \cos \alpha} = \tan \alpha - \frac{gt}{2V \cos \alpha}$	Hence $\tan \theta = \tan \alpha - \frac{gt}{2V \cos \alpha}$ At Q, $\dot{x} = V \cos \alpha$ $\frac{dy}{dx} = \frac{V \sin \alpha - gt}{V \cos \alpha}$ $\tan \beta = \tan \alpha - \frac{gt}{V \cos \alpha}$ so $\frac{gt}{V \cos \alpha} = \tan \alpha - \tan \beta$ Hence $\tan \theta = \tan \alpha - \frac{\tan \alpha - \tan \beta}{2}$ $2 \tan \theta = 2 \tan \alpha - \tan \alpha - \tan \beta$ $\tan \beta = 2 \tan \theta - \tan \alpha$ $\beta = \tan^{-1}(2 \tan \theta - \tan \alpha)$	1 Gradient of PQ
Hence $\tan \theta = \tan \alpha - \frac{gt}{2V \cos \alpha}$	At Q, $\dot{x} = V \cos \alpha$	1 $\dot{x}$ and y if nothing else
$\frac{dy}{dx} = \frac{V \sin \alpha - gt}{V \cos \alpha}$	$\tan \beta = \tan \alpha - \frac{gt}{V \cos \alpha}$	1 for $\frac{dy}{dx}$
so $\frac{gt}{V \cos \alpha} = \tan \alpha - \tan \beta$	Hence $\tan \theta = \tan \alpha - \frac{\tan \alpha - \tan \beta}{2}$	1 for $\tan \beta$
$2 \tan \theta = 2 \tan \alpha - \tan \alpha - \tan \beta$	$\tan \beta = 2 \tan \theta - \tan \alpha$	1 evidence of correct simplification to given result

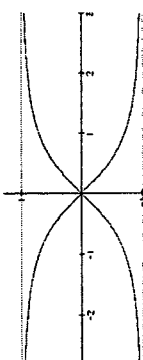
Question 8

(a) (i) $I_n = \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \csc^n x \, dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \csc^{n-2} x \csc^2 x \, dx$ $= -\csc^{n-2} x \cot x \Big _{\frac{\pi}{6}}^{\frac{\pi}{2}} - \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (n-2) \csc^{n-3} x (-1) (\sin x)^{-2} \cos x (-\cot x) \, dx$ $= 0 + 2^{n-2} \times \sqrt{3} - (n-2) \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \csc^n x \cos^2 x \, dx$ $= 2^{n-2} \times \sqrt{3} - (n-2) \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \csc^n x (1 - \sin^2 x) \, dx$ $= 2^{n-2} \times \sqrt{3} - (n-2) \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \csc^n x \, dx + (n-2) \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \csc^n x \sin^2 x \, dx$ $= 2^{n-2} \times \sqrt{3} - (n-2) I_n + (n-2) I_{n-2}$ $I_n + (n-2) I_n = 2^{n-2} \times \sqrt{3} + (n-2) I_{n-2}$ $(n-1) I_n = 2^{n-2} \times \sqrt{3} + (n-2) I_{n-2}$ (ii) $\sec x = \csc \left(\frac{\pi}{2} - x \right)$ $y = \frac{\pi}{2} - x \Rightarrow dy = -dx$	4
$J = \int_0^{\frac{\pi}{3}} \sec^4 x \, dx = \int_0^{\frac{\pi}{3}} \sec^2 x \sec^2 x \, dx$ $= \int_0^{\frac{\pi}{3}} \sec^2 x (1 + \tan^2 x) \, dx$ OR $J = \int_0^{\frac{\pi}{3}} \sec^4 x \, dy$ $= \left[\tan x + \frac{\tan^3 x}{3} \right]_0^{\frac{\pi}{3}}$ $= \sqrt{3} + \frac{3\sqrt{3}}{3} - 0 = 2\sqrt{3}$	1 correct evaluation of uv
$J = \int_0^{\frac{\pi}{3}} \sec^4 x \left(\frac{\pi}{2} - x \right) dx$ $= \int_0^{\frac{\pi}{3}} \sec^4 x \, y (-dy)$ $= -\int_0^{\frac{\pi}{3}} \sec^4 x \, y \, dy$ $= -\left[\frac{\sec^4 x \, y^2}{2} - \int \frac{2 \sec^4 x \, y}{2} dy \right]_0^{\frac{\pi}{3}}$ $= -\left[\frac{\sec^4 x \, y^2}{2} - \sec^4 x \, y \right]_0^{\frac{\pi}{3}}$ $= -\left[\frac{16}{2} \cdot \frac{9}{4} - 16 \cdot \frac{3}{2} \right] = -\left[36 - 24 \right] = -12$	1
Method 1 1 correct integrand 1 correct primitive 1 correct substitution	3
Method 2 Need evidence of change of variable, change of limits before using result from (i) $3I_4 = 4\sqrt{3} + 2I_2$ $I_2 = 2^0 \sqrt{3} + 0$ $= \sqrt{3}$ $3I_4 = 4\sqrt{3} + \sqrt{3} = 5\sqrt{3}$ $I_4 = \frac{5\sqrt{3}}{3}$	Method 2

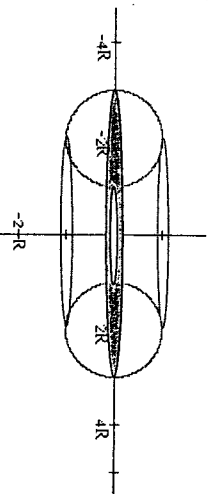
Question 7

(a) (i) N is $\left(\frac{3p+3q}{2}, \frac{p}{2} \right)$ ie $\left(\frac{3(p+q)}{2}, \frac{3(p+q)}{2pq} \right)$ (ii) Since PQ is a tangent to the parabola $y^2 = 3x$, the quadratic equation obtained by solving simultaneously with $x + pqy = 3(p+q)$ will have a double root. $y^2 = 3(3(p+q) - pqy)$ $y^2 + 3pqy - 9(p+q) = 0$ $\therefore (3pq)^2 - 4 \times 1 \times [-9(p+q)] = 0$ $(3pq)^2 + 36(p+q) = 0$ $(pq)^2 + 4(p+q) = 0$ OR From N, $x = \frac{3(p+q)}{2}$, $y = \frac{3(p+q)}{2pq}$ ie. $p+q = \frac{2x}{3}$, $pq = \frac{x}{y}$ so $\left(\frac{3x}{y} \right)^2 + 36 \times \frac{2x}{3} = 0$ $\frac{9x^2}{y^2} = -24x$ $\frac{x^2}{y^2} = -8x$ $3x = -8y^2$ $3x = -8y^2$ is the locus of N. NB: $\left(y + \frac{3pq}{2} \right)^2 = \frac{9p^2q^2}{4} + 9(p+q)$ And Gradient of $PQ = \frac{-1}{pq}$ $\frac{dy}{dx} = \frac{3}{-2y}$ $\therefore -2y = 3pq$	1 3 1 simplification of Δ 1 expressions for $p+q$ and pq from N 1 for $\left(\frac{3x}{y} \right)^2 + 36 \times \frac{2x}{3} = 0$ or equivalent Pay 1 Pay 1
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Question 7 continued

(b) (i) $x^2y^2 - x^2 + y^2 = 0$ $x^2y^2 + y^2 = x^2$ $y^2(x^2 + 1) = x^2$ $y^2 = \frac{x^2}{x^2 + 1}$ $y^2 = 1 - \frac{1}{x^2 + 1}$ As $x \rightarrow \infty$, $y^2 \rightarrow 1$ from below and hence $y < 1$ (ii) Find the equations of the horizontal asymptotes. $y^2 = 1 - \frac{1}{x^2 + 1}$ As $x \rightarrow \infty$, $y^2 \rightarrow 1$ Hence equations of horizontal asymptotes are $y = \pm 1$ (iii) $x^2y^2 - x^2 + y^2 = 0$ $2xy^2 + x^2 \times 2y \frac{dy}{dx} - 2x + 2y \frac{dy}{dx} = 0$ $y(x^2 + 1) \frac{dy}{dx} = x(1 - y^2)$ $\Rightarrow \frac{dy}{dx} = \frac{x(1 - y^2)}{y(x^2 + 1)}$ From (i), $x^2(y^2 - 1) + y^2 = 0 \Rightarrow 1 - y^2 = \frac{y^2}{x^2}$ $y^2(x^2 + 1) - x^2 = 0 \Rightarrow 1 + x^2 = \frac{x^2}{y^2}$ $\frac{dy}{dx} = \frac{x}{y} \times \frac{y^2}{x^2} \times \frac{y^2}{x^2} = \frac{y^3}{x^3}$ (iv) 	2 1 1 1 3 1 evidence of partially correct implicit differentiation 1 correct expression for $\frac{dy}{dx}$ 1 using these results or equivalent to obtain correct answer 1 based on evidence in earlier parts
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(b)



$$(x-2R)^2 + y^2 = R^2$$

$$y^2 = R^2 - (x-2R)^2$$

$$y = \pm \sqrt{R^2 - (x-2R)^2}$$

$y = \sqrt{R^2 - (x-2R)^2}$ is upper boundary.

$$\delta V = 2\pi x \times 2y \times \delta x$$

$$V = \int_R^{3R} 4\pi xy dx$$

$$= \int_R^{3R} 4\pi x \sqrt{R^2 - (x-2R)^2} dx$$

Let $x-2R = R \sin \theta$

$$x = R, \quad \theta = -\frac{\pi}{2}$$

$$x = 3R, \quad \theta = \frac{\pi}{2}$$

$$dx = R \cos \theta d\theta$$

$$V = 4\pi \int_{-\pi/2}^{\pi/2} (2R + R \sin \theta) \sqrt{R^2 - R^2 \sin^2 \theta} R \cos \theta d\theta$$

$$= 4\pi \int_{-\pi/2}^{\pi/2} (2R + R \sin \theta) R^2 \cos^2 \theta d\theta$$

$$= 4\pi R^3 \int_{-\pi/2}^{\pi/2} (2 \cos^2 \theta + \sin \theta \cos^2 \theta) d\theta$$

$$= 4\pi R^3 \left[\theta + \sin 2\theta - \frac{\cos^3 \theta}{3} \right]_{-\pi/2}^{\pi/2}$$

$$= 4\pi R^3 \left(\frac{\pi}{2} + 0 - 0 - \left(-\frac{\pi}{2} + 0 - 0 \right) \right)$$

$$= 4\pi^2 R^3 \text{ units}^3$$

6

1 equ'n of upper boundary

1 correct definite integral for V

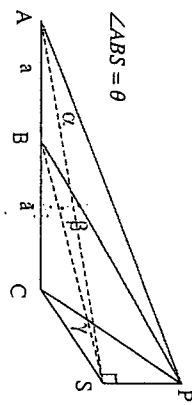
1 correct substitution for y

1 set up correct substitution with new limits

1 simplified integrand

1 correct primitive with correct limits

(c)



(i) $\angle ABS = \theta$ hence $\angle CBS = 180 - \theta$

$$\text{In } \triangle CBS, CS^2 = BC^2 + BS^2 - 2 \times BC \times BS \cos(180 - \theta)$$

$$\therefore CS^2 = a^2 + BS^2 - 2 \times a \times BS \cos \theta$$

$$\text{In } \triangle BSP, BS = h \cot \beta$$

$$\therefore CS^2 = a^2 + h^2 \cot^2 \beta + 2ah \cot \beta \cos \theta$$

(ii)

$$\text{In } \triangle ASP, AS = h \cot \alpha$$

$$\text{In } \triangle CSP, CS = h \cot \gamma$$

$$\therefore h^2 \cot^2 \gamma = a^2 + h^2 \cot^2 \beta + 2ah \cot \beta \cos \theta$$

$$\text{In } \triangle ABS, AS^2 = AB^2 + BS^2 - 2 \times AB \times BS \cos \theta$$

$$\therefore h^2 \cot^2 \alpha = a^2 + h^2 \cot^2 \beta - 2ah \cot \beta \cos \theta$$

$$\text{Hence } \cos \theta = \frac{a^2 + h^2 \cot^2 \beta - \cot^2 \alpha}{2ah \cot \beta}$$

$$\text{But } h^2 \cot^2 \gamma = a^2 + h^2 \cot^2 \beta + 2ah \cot \beta \cos \theta \text{ from (i)}$$

$$\text{So } \cos \theta = \frac{h^2 (\cot^2 \gamma - \cot^2 \beta) - a^2}{2ah \cot \beta}$$

$$\text{Hence } \frac{a^2 + h^2 (\cot^2 \beta - \cot^2 \alpha)}{2ah \cot \beta} = \frac{h^2 (\cot^2 \gamma - \cot^2 \beta) - a^2}{2ah \cot \beta}$$

$$a^2 + h^2 (\cot^2 \beta - \cot^2 \alpha) = h^2 (\cot^2 \gamma - \cot^2 \beta) - a^2$$

$$2a^2 = h^2 (\cot^2 \gamma - \cot^2 \beta - \cot^2 \beta + \cot^2 \alpha)$$

$$h^2 = \frac{2a^2}{\cot^2 \gamma - 2\cot^2 \beta + \cot^2 \alpha}$$

$$h = \frac{a\sqrt{2}}{\sqrt{\cot^2 \alpha + \cot^2 \gamma - 2\cot^2 \beta}}$$

2

1 correct cos rule with evidence of $180 - \theta$

1 change of sign and substitution for BS

3

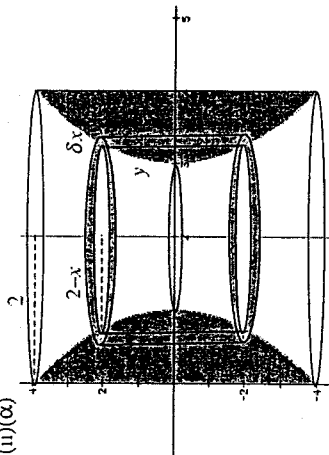
1 expression for $\cos \theta$

1 second expression for $\cos \theta$

1 correct substitution and simplification to expression for $2a^2$

Question 5 continued

(ii)(α)



$$\begin{aligned}\delta V &= 2\pi(2-x) \times 2y \times \delta x \\ V &= 4\pi \int_0^1 (2-x)y \, dx \\ \text{but } y^2 &= 16(1-x) \\ \text{Use } y &= 4\sqrt{1-x} \text{ as upper branch} \quad \text{OR} \\ V &= 4\pi \int_0^1 (2-x)4\sqrt{1-x} \, dx \\ &= 16\pi \int_0^1 (2-x)\sqrt{1-x} \, dx \\ R-r &= 2-x+\delta x-2+x \\ &= \delta x \\ R+r &= 2-x+\delta x+2-x \\ &= 4-2x+\delta x \\ (R-r)(R+r) &= \delta x(4-2x+\delta x) \\ &= 2(2-x)\delta x \\ \delta V &= \pi(R^2-r^2) \times 2y \\ &= \pi \times 2(2-x)\delta x \times 2y \\ &= 4\pi(2-x)y\delta x\end{aligned}$$

(β)

$$\begin{aligned}&\int_0^1 16\pi(2-x)\sqrt{1-x} \, dx \quad \text{Let } u = 1-x, \quad du = -dx \\ &= \int_1^0 -16\pi(1+u)\sqrt{u} \, du \\ &= 16\pi \int_0^1 (\sqrt{u} + u^{3/2}) \, du \\ &= 16\pi \left[\frac{2}{3}u^{3/2} + \frac{2}{5}u^{5/2} \right]_0^1 \\ &= 16\pi \left(\frac{2}{3} + \frac{2}{5} \right) = \frac{256\pi}{15}\end{aligned}$$

3

1 evidence of correct substitution
1 correct primitive
1 correct substitution
2 for $\frac{-256\pi}{15}$ with working

2

Evidence of $y = 4\sqrt{1-x}$ is needed for 1

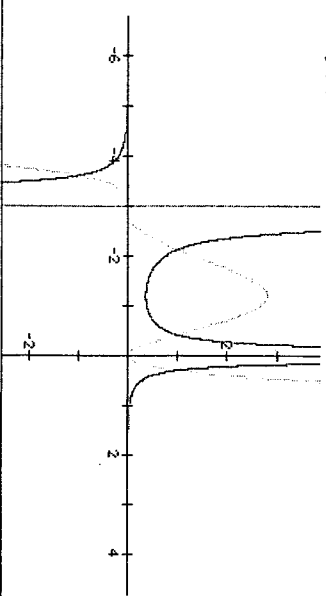
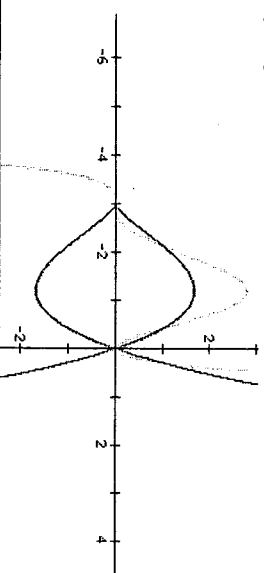
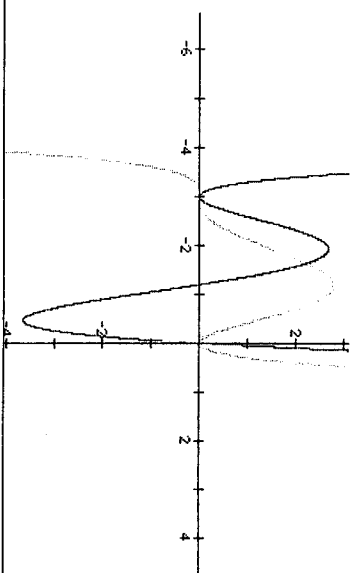
Question 6

(a)

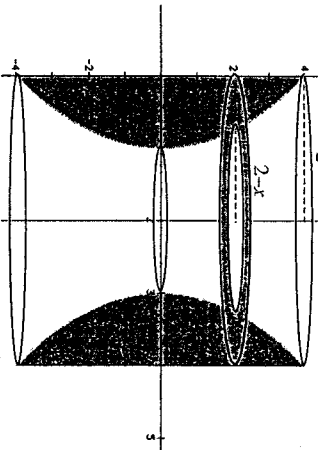
Base of each triangle is $2y$
Area of triangle is $\frac{1}{2} \times 2y \times 2y \times \frac{\sqrt{3}}{2} = \sqrt{3}y^2$.
 $\delta V = \sqrt{3}y^2 \delta x$
 $V = \int_{-6}^6 \sqrt{3}y^2 \, dx$ and $y^2 = \frac{4}{9}(36-x^2)$
 $= 2 \int_0^6 \sqrt{3} \times \frac{4}{9}(36-x^2) \, dx$
 $= \frac{8\sqrt{3}}{9} \int_0^6 (36-x^2) \, dx = \frac{8\sqrt{3}}{9} \left[36x - \frac{x^3}{3} \right]_0^6$
 $= \frac{8\sqrt{3}}{9} (216 - 72) = 128\sqrt{3} \text{ units}^3$

4

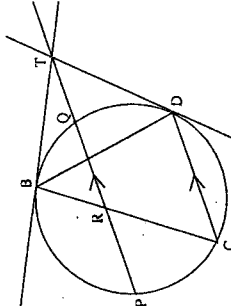
1 area of Δ
1 indefinite integral + expression for y^2
OR definite integral with correct limits
1 correct integral prior to integration
1 correct primitive + correct subst shown

(a)(i) $y = \frac{1}{f(x)}$		2 1 for two parts correct
(ii) $y^2 = f(x)$		2 Need evidence of graph becoming horizontal at $x = -3$ 1 for half correct
(iii) $y = f'(x)$		1
(b) $f(x) = kx^2(x+3)^3$ Also accept $f(x) = kx^4(x+3)^3$		1

Question 5 continued

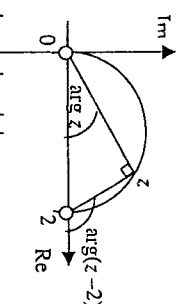
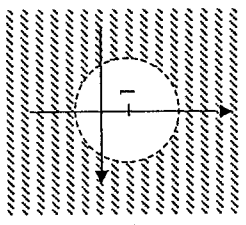
(c)		4
(i)	$\delta V = \pi(2^2 - (2-x)^2)\delta y$ $\delta V = \pi(2 - 2 + x)(2 + 2 - x)\delta y$ $\delta V = \pi x(4 - x)\delta y$ $V = \int_{-4}^4 \pi x(4 - x) dy \quad \text{but} \quad x = 1 - \frac{y^2}{16} \quad \text{or} \quad x = \frac{16 - y^2}{16}$ $= \pi \int_{-4}^4 \left(1 - \frac{y^2}{16}\right) \left(4 - 1 + \frac{y^2}{16}\right) dy$ $= \pi \int_{-4}^4 \left(3 - \frac{y^2}{8} - \frac{y^4}{256}\right) dy$ $= 2\pi \left[3y - \frac{y^3}{24} - \frac{y^5}{1280} \right]_{-4}^4 = 2\pi \left(12 - \frac{64}{24} - \frac{4^5 \times 4}{256 \times 5} - 0 \right) = \frac{256\pi}{15} \text{ units}^3$	4 1 for δV correct Evidence of the use of a correct substitution for x 1 correct primitive 1 correct substitution 3 for $\frac{416\pi}{15}$ with working

Question 4 continued

(d) (i) $2\alpha, 2\beta, 2\gamma$ $\left(\frac{x}{2}\right)^3 + 2\left(\frac{x}{2}\right)^2 - 2\left(\frac{x}{2}\right) + 3 = 0$ $x^3 + 4x^2 - 8x + 24 = 0$ OR $\frac{x^3}{8} + \frac{x^2}{2} - x + 3 = 0$ (ii) $\alpha^2, \beta^2, \gamma^2$ $(\sqrt{x})^3 + 2(\sqrt{x})^2 - 2(\sqrt{x}) + 3 = 0$ $x\sqrt{x} + 2x - 2\sqrt{x} + 3 = 0$ $\sqrt{x}(x-2) = -(2x+3)$ $x(x-2)^2 = (2x+3)^2$ $x^3 - 4x^2 + 4x = 4x^2 + 12x + 9$ $x^3 - 8x^2 - 8x - 9 = 0$	1 3 1 this line 1 this line 1 this line
(e) 	2 2 (i) $\angle BDT = \angle BCD$ (angle between tangent and chord equals angle in alternate segment) $\angle BRT = \angle BCD$ (corresponding angles equal, $RQ \parallel CD$) $\therefore \angle BDT = \angle BRT$ (ii) $\angle BDT = \angle BRT$ from (i). These are a pair of equal angles standing on the same side of BT. Hence B, T, D and R are concyclic points

(a) (i) Factorise $P(x) = x^4 - 5x^3 + 4x^2 + 2x - 8$ over reals $P(x) = x^4 - 5x^3 + 4x^2 + 2x - 8$ $P(-1) = 1 + 5 + 4 - 2 - 8 = 0$ $(x+1)$ is a factor. $\begin{array}{r} x^3 - 6x^2 + 10x - 8 \\ x^4 - 5x^3 + 4x^2 + 2x - 8 \\ \hline x^4 + x^3 \\ \hline -6x^3 + 4x^2 \\ \hline -6x^3 + 4x^2 \\ \hline 10x^2 + 2x \\ \hline 10x^2 + 10x \\ \hline -8x - 8 \\ \hline -8x - 8 \\ \hline 0 \end{array}$ $Q(x) = x^3 - 6x^2 + 10x - 8$ $Q(4) = 64 - 96 + 40 - 8 = 0$ $(x-4)$ is a factor $\begin{array}{r} x^2 - 2x + 2 \\ x^3 - 6x^2 + 10x - 8 \\ \hline x^3 - 4x^2 \\ \hline -2x^2 + 10x \\ \hline -2x^2 + 8x \\ \hline 2x - 8 \\ \hline 2x - 8 \\ \hline 0 \end{array}$ $P(x) = (x+1)(x-4)(x^2 - 2x + 2) \text{ over the reals}$ (ii) $P(x) = (x+1)(x-4)(x^2 - 2x + 2)$ $x^2 - 2x + 2 = (x-1)^2 + 1$ $= (x-1+i)(x-1-i)$ $P(x) = (x+1)(x-4)(x-1+i)(x-1-i) \text{ over the complex numbers.}$	2 1 for one factor $x = \frac{2 \pm \sqrt{4-4 \times 1 \times 2}}{2}$ $= \frac{2 \pm \sqrt{-4}}{2}$ $= 1 \pm i$ $(x-(1+i))(x-(1-i))$ $(x-1-i)(x-1+i)$ 1
(b) $P(x) = k(x-3)(x+1)^3$ (c)(i) $x^3 - 2x^2 + x + 3 = 0$ $\alpha + \beta + \gamma = 2 \quad \alpha\beta + \beta\gamma + \gamma\alpha = 1 \quad \alpha\beta\gamma = -3$ $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$ $= 4 - 2 = 2$ (ii) $\alpha^3 - 2\alpha^2 + \alpha + 3 = 0$ $\beta^3 - 2\beta^2 + \beta + 3 = 0$ $\gamma^3 - 2\gamma^2 + \gamma + 3 = 0$ $\alpha^3 + \beta^3 + \gamma^3 - 2(\alpha^2 + \beta^2 + \gamma^2) + \alpha + \beta + \gamma + 9 = 0$ $\alpha^3 + \beta^3 + \gamma^3 - 4 + 2 + 9 = 0$ $\alpha^3 + \beta^3 + \gamma^3 = -7$	1 2 1

Question 3

(a) $w = -1 - 3i, \quad \bar{w} = -1 + 3i$ $\arg((2+i)w) = \arg((2+i)(-1+3i))$ $= \arg(-2 - i + 6i - 3)$ $= \arg(-5 + 5i)$ $= \frac{3\pi}{4}$	2 1 for $-\frac{\pi}{4}$ or $\frac{\pi}{4}$.
(b) $x^2 - 12x + 48 = (x-6)^2 + 12$ $= (x-6 + i\sqrt{12})(x-6 - i\sqrt{12})$ $= (x-6 + 2\sqrt{3}i)(x-6 - 2\sqrt{3}i)$	2 $x = \frac{12 \pm \sqrt{144 - 192}}{2}$ $= \frac{12 \pm \sqrt{48}i}{2}$ $= 6 \pm 2\sqrt{3}i$
(c) OQ is just OP rotated clockwise through an angle of $\frac{\pi}{2}$. Q is represented by $-i(a+bi) = b - ai$	1
(d)(i) $\arg(z-2) = \arg z + \frac{\pi}{2}$ may be written $\arg(z-2) - \arg z = \frac{\pi}{2}$ which suggests an angle in a semi circle, centre $(1, 0)$ radius 1 in the upper half plane, excluding the points $(0, 0)$ and $(2, 0)$.  (ii) $ z+3i < 2 z $ Let $z = x + iy$ $ x + (3+y)i < 2 x + iy $ $x^2 + (y+3)^2 < 4(x^2 + y^2)$ $x^2 + y^2 + 6y + 9 < 4x^2 + 4y^2$ $3x^2 + 3y^2 - 6y > 9$ $x^2 + y^2 - 2y > 3$ $x^2 + (y-1)^2 > 4$ This region is the exterior of the circle centre $(0, 1)$, radius 2	2 1 1 3 1 for dotted circle -1 for wrong centre 

Question 3 continued

(e) (i) Let $(r \operatorname{cis} \theta)^3 = -8$ $r^3 \operatorname{cis} 3\theta = -8$ $r = 2, \quad \operatorname{cis} 3\theta = -1$ $3\theta = \pi, \pi + 2\pi, \pi - 2\pi$ $\theta = \frac{\pi}{3}, \pi, \frac{-\pi}{3}$ The three cube roots of -8 are $2 \operatorname{cis} \left(\frac{-\pi}{3} \right), 2 \operatorname{cis} \frac{\pi}{3}, 2 \operatorname{cis} \pi$ or $-2 \operatorname{cis} \left(\frac{\pm 2\pi}{3} \right), 2 \operatorname{cis} \pi$	2
(ii) $2 \operatorname{cis} \left(\frac{-\pi}{3} \right) = 2 \left(\cos \frac{-\pi}{3} + i \sin \frac{-\pi}{3} \right) = 2 \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i \right) = 1 - \sqrt{3}i$ $2 \operatorname{cis} \frac{\pi}{3} = 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) = 2 \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) = 1 + \sqrt{3}i$	1
(iii) Let $w_1 = 2 \operatorname{cis} \left(\frac{-\pi}{3} \right), w_2 = 2 \operatorname{cis} \frac{\pi}{3}$ $w_1^{6n} + w_2^{6n} = \left(2 \operatorname{cis} \left(\frac{-\pi}{3} \right) \right)^{6n} + \left(2 \operatorname{cis} \left(\frac{\pi}{3} \right) \right)^{6n}$ $= 2^{6n} (\operatorname{cis}(-2\pi n) + \operatorname{cis}(2\pi n))$ $= 2^{6n} (\cos(-2\pi n) + i \sin(-2\pi n) + \cos(2\pi n) + i \sin(2\pi n))$ $= 2^{6n} (1 + 0 + 1 + 0) \text{ when } n \text{ is an integer}$ $= 2 \times 2^{6n}$ $= 2^{6n+1}$	2 1 Induction with everything set up correctly with an error
OR $w_1^3 = -8, w_2^3 = -8$ $w_1^{6n} + w_2^{6n} = (-8)^{2n} + (-8)^{2n}$ $= (-2)^{6n} + (-2)^{6n}$ $= 2 \times (-2)^{6n}$ $= 2 \times 2^{6n}$ $= 2^{6n+1}$	

Question 2

(a)(i) $\int \frac{x}{\sqrt{9-16x^2}} dx = \frac{1}{16} \int \frac{16x}{\sqrt{9-16x^2}} dx = \frac{-1}{16} \sqrt{9-16x^2} + C$ OR $x = \frac{3}{4} \sin \theta, \quad dx = \frac{3}{4} \cos \theta d\theta$ $\int \frac{x}{\sqrt{9-16x^2}} dx = \int \frac{\frac{3}{4} \sin \theta \times \frac{3}{4} \cos \theta d\theta}{\sqrt{9-16 \times \frac{9}{16} \sin^2 \theta}} = \frac{9}{16} \int \frac{\sin \theta \cos \theta d\theta}{\sqrt{9-9 \sin^2 \theta}}$ $= \frac{3}{16} \int \sin \theta d\theta = \frac{-3}{16} \cos \theta + C$ $= \frac{-3}{16} \sqrt{1-\frac{4}{9}x^2} + C = \frac{-1}{16} \sqrt{9-4x^2} + C$ (ii) $\int \frac{x^2}{x+1} dx = \int \frac{(x+1)(x-1)+1}{x+1} dx$ $= \int \left(x-1 + \frac{1}{x+1} \right) dx = \frac{x^2}{2} - x + \ln x+1 + C$ (iii) $\int_0^{\ln 3} x e^x dx = x e^x \Big _0^{\ln 3} - \int_0^{\ln 3} e^x dx$ $= 3 \ln 3 - [e^x]_0^{\ln 3} = 3 \ln 3 - 3 + 1 = 3 \ln 3 - 2$	2 3 1 for first line 1 for indefinite integral 1 for correct substit 3 -1 for error carried through (b)(i) $\frac{2}{(t+1)(t^2+1)} = \frac{A}{t+1} + \frac{Bt+C}{t^2+1}$ $2 \equiv A(t^2+1) + (Bt+C)(t+1)$ $t = -1 \Rightarrow 2 = 2A \quad \therefore A = 1$ $t = 0 \Rightarrow 2 = A + C \quad \therefore C = 1$ $t = 1 \Rightarrow 2 = 2 + (B+1) \times 2 \quad \therefore B = -1$
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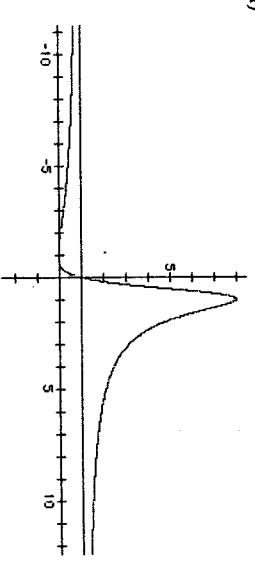
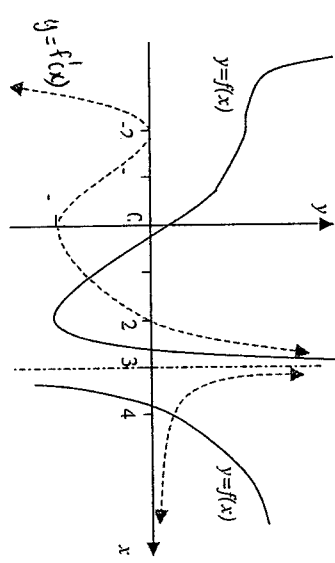
Question 2 continued

(b)(ii) $\int_0^1 \frac{2}{(t+1)(t^2+1)} dt = \int_0^1 \left(\frac{1}{(t+1)} + \frac{1-t}{(t^2+1)} \right) dt$ $= \int_0^1 \left(\frac{1}{(t+1)} + \frac{1}{(t^2+1)} - \frac{t}{(t^2+1)} \right) dt$ $= \left[\ln t+1 + \tan^{-1} t - \frac{1}{2} \ln(t^2+1) \right]_0^1$ $= \left[\tan^{-1} t + \ln \frac{ t+1 }{\sqrt{t^2+1}} \right]_0^1$ $= \tan^{-1} 1 + \ln \frac{2}{\sqrt{2}} - \left(0 + \ln \frac{1}{\sqrt{1}} \right)$ $= \frac{\pi}{4} + \frac{1}{2} \ln 2$	3 2 1 for mostly correct 1 for this integral No mark for last line (iii) $t = \tan \frac{x}{2} \Rightarrow dx = \frac{2dt}{1+t^2}, \quad x=0, t=0 \quad x=\frac{\pi}{2}, t=1$ $\int_0^{\frac{\pi}{2}} \frac{\sin x}{1+\sin x - \cos x} dx = \int_0^1 \frac{\frac{2t}{1+t^2} \times \frac{2dt}{1+t^2}}{1 + \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2}}$ $= \int_0^1 \frac{4t dt}{(1+t^2)(1+t^2+2t-1+t^2)}$ $= \int_0^1 \frac{4t dt}{(1+t^2)(2t^2+2t)}$ $= \int_0^1 \frac{2 dt}{(1+t^2)(t+1)}$ $= \frac{\pi}{4} + \frac{\ln 2}{2}$ from b(ii) $\Rightarrow$
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Question 1

(a) If $f(x) = (x-1)(x-3)$ then sketch (i) $y = \frac{1}{f(x)}$	2 1 for both top branches 1 for bottom part with correct local max
(ii) $y = f(x)$	2 1 for one half correct
(iii) $ y = f(x)$	2 1 for either vertical half or either horizontal half

Question 1 continued

(b) (i) $y = \frac{(x+1)^4}{x^4+1}$ $\frac{dy}{dx} = \frac{4(x+1)^3(x^4+1) - (x+1)^4 \times 4x^3}{(x^4+1)^2}$ $= \frac{4(x+1)^3(x^4+1-x^4-x^3)}{(x^4+1)^2} = \frac{4(x+1)^3(1-x^3)}{(x^4+1)^2} = 0$ when $x = -1$ or $x = 1$. Stationary points are $(-1, 0)$, $(1, 8)$ For asymptotes, write $y = \frac{(x+1)^4}{x^4+1} = \frac{x^4(1+\frac{1}{x})^4}{x^4+1} = \frac{(1+\frac{1}{x})^4}{1+\frac{1}{x^4}}$ As $x \rightarrow \pm\infty$, $y = \frac{(1+\frac{1}{x})^4}{1+\frac{1}{x^4}} \rightarrow 1$, horizontal asymptote is $y = 1$.	2 1 for stationary points 1 for asymptotes
(ii) 	1 if correct and not made easier from errors above
(iii) $y = \frac{(x+1)^4}{x^4+1}$ and $y = k \rightarrow k(x^4+1) = (x+1)^4$ Two roots if $0 < k < 1$ or $1 < k < 8$, from the graph.	2 1 for each set of values 1 for $0 < k < 8$
(c) 	4 deduct 1 for each part missing. horizontal at $x = -2$ slope -1 at $x = 0$ minimum at $x = 2$ asymptotic to $x = 3$ RH branch monotonic increase & flattening